

Algebraic Identities

Three identities that come straight from the distributive property — and that turn hard squarings and products into quick mental maths.

The three identities

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a - b)^2 = a^2 - 2ab + b^2$$

$$(a + b)(a - b) = a^2 - b^2$$

Each comes from multiplying out the brackets. The only difference between the first two is the sign of the middle term.

Using them for fast calculation

- Split a number into an easy sum or difference. $55^2 = (50 + 5)^2 = 2500 + 500 + 25 = 3025$.
- $99^2 = (100 - 1)^2 = 10000 - 200 + 1 = 9801$.
- $397 \times 403 = (400 - 3)(400 + 3) = 160000 - 9 = 159991$.

Bonus identities: $2(a^2 + b^2) = (a + b)^2 + (a - b)^2$, and $a^2 = (a + b)(a - b) + b^2$ — the second is a neat trick for squaring (e.g. $31^2 = 32 \times 30 + 1 = 961$).

Prerequisite refresher: "ab" means $a \times b$, and the "middle term" $2ab$ is twice the product of the two parts.

Where marks slip away

Trap 1 — Don't forget the middle term. $(a + b)^2 = a^2 + 2ab + b^2$, NOT $a^2 + b^2$. The $2ab$ is the most-missed term.

Trap 2 — Sign of the middle term. In $(a - b)^2$ it is $-2ab$; in $(a + b)^2$ it is $+2ab$.

Trap 3 — Square each whole part. In $(6x + 5)^2$, $(6x)^2 = 36x^2$, not $6x^2$.

Trap 4 — Difference of squares needs $a - b$. $(a + b)(a - b) = a^2 - b^2$ only when the two brackets are a sum and the matching difference.

Slip 5 — $(a - b)^2 = (b - a)^2$. Squaring removes the sign, so both give the same value.

Model answers — write them like this

Example A. Expand $(6x + 5)^2$.

Step 1: Use $(a + b)^2$ with $a = 6x$, $b = 5$.

Step 2: $(6x)^2 + 2(6x)(5) + 5^2 = 36x^2 + 60x + 25$.

$$\therefore 36x^2 + 60x + 25$$

Example B. Expand $(3j + 2k)^2$.

Step 1: $a = 3j$, $b = 2k$.

Step 2: $(3j)^2 + 2(3j)(2k) + (2k)^2 = 9j^2 + 12jk + 4k^2$.

$$\therefore 9j^2 + 12jk + 4k^2$$

Example C. Find 99^2 using an identity.

Step 1: $99 = 100 - 1$, so use $(a - b)^2$ with $a = 100$, $b = 1$.

Step 2: $100^2 - 2(100)(1) + 1^2 = 10000 - 200 + 1$.

$$\therefore 99^2 = 9801$$

Example D. Find 397×403 using an identity.

Step 1: $397 = 400 - 3$ and $403 = 400 + 3$, so use $(a - b)(a + b) = a^2 - b^2$.

Step 2: $400^2 - 3^2 = 160000 - 9$.

$$\therefore 397 \times 403 = 159991$$

Next: [try the worksheet](#), then [check the answer key](#).